

Hierarchical Inheritance in Java

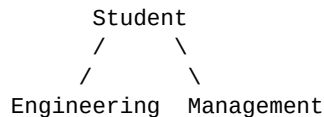
Java Programming Lab Record — Multiple Test Cases

1. Aim

To implement Hierarchical Inheritance in Java and execute the program using multiple test cases to verify the results.

2. Concept

In Hierarchical Inheritance, multiple child classes inherit from a single parent class.



- **Student** → Parent class
- **Engineering** → Child class
- **Management** → Child class

3. Java Program

```
import java.util.Scanner;

// Parent class
class Student {
    String name;
    int rollNo;
    int marks;

    void getStudentDetails(String name, int rollNo, int marks) {
        this.name = name;
        this.rollNo = rollNo;
        this.marks = marks;
    }

    void displayStudentDetails() {
        System.out.println("Student Name: " + name);
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Marks: " + marks);
    }
}

// Child class 1
class Engineering extends Student {

    void displayEngineeringResult() {
        displayStudentDetails();

        if (marks >= 35) {
            System.out.println("Engineering Result: Pass");
        } else {
            System.out.println("Engineering Result: Fail");
        }
    }
}

// Child class 2
class Management extends Student {
    void displayManagementResult() {
```

```

        displayStudentDetails();

        if (marks >= 35) {
            System.out.println("Management Result: Pass");
        } else {
            System.out.println("Management Result: Fail");
        }
    }
}

// Main class
public class HierarchicalInheritance {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of test cases: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 1; i <= n; i++) {

            System.out.println("\n----- Test Case " + i + " -----");

            System.out.print("Enter student name: ");
            String name = sc.nextLine();

            System.out.print("Enter roll number: ");
            int rollNo = sc.nextInt();

            System.out.print("Enter marks: ");
            int marks = sc.nextInt();

            System.out.print("Enter course (1-Engineering, 2-Management): ");
            int course = sc.nextInt();
            sc.nextLine();

            if (course == 1) {

                // Object of Engineering child class
                Engineering e = new Engineering();

                e.getStudentDetails(name, rollNo, marks);
                e.displayEngineeringResult();

            } else if (course == 2) {

                // Object of Management child class
                Management m = new Management();

                m.getStudentDetails(name, rollNo, marks);
                m.displayManagementResult();

            } else {

                System.out.println("Invalid course!");
            }
        }

        sc.close();
    }
}

```

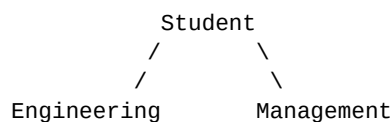
4. Explanation

- **Student** is the single parent class, holding name, roll number, and marks, plus a shared `displayStudentDetails()` method.
- **Engineering** and **Management** are two separate child classes that both extend `Student` independently — neither inherits from the other.
- Each child class reuses the inherited fields and `displayStudentDetails()` method, and adds its own result-checking method.
- Based on the course chosen at runtime, the program creates an `Engineering` or `Management` object and calls the matching result method.

5. Test Case Verification

Test Case	Name	Course	Marks	Expected	Actual
1	hin	Engineering	78	Pass	Pass ■
2	furt	Management	87	Pass	Pass ■

6. Inheritance Structure



Why is this Hierarchical Inheritance? Because two different child classes (`Engineering` and `Management`) inherit from the same parent class (`Student`).

Note: Unlike multilevel inheritance, there is no chain such as $A \rightarrow B \rightarrow C$. Instead, it is one parent $\rightarrow$ multiple children.

7. Output Screenshot

The screenshot below shows the program actually executed, running both test cases in a single session:

```
<terminated> HierarchicalInheritance [Java Application] C:\Users\logid\De
Enter number of test cases: 2

----- Test Case 1 -----
Enter student name: hin
Enter roll number: 101
Enter marks: 78
Enter course (1-Engineering, 2-Management): 1
Student Name: hin
Roll Number: 101
Marks: 78
Engineering Result: Pass

----- Test Case 2 -----
Enter student name: furt
Enter roll number: 321
Enter marks: 87
Enter course (1-Engineering, 2-Management): 2
Student Name: furt
Roll Number: 321
Marks: 87
Management Result: Pass
```

The results verify that both child classes correctly reuse the inherited fields and the shared display method from Student, confirming hierarchical inheritance works as expected.